

Comparative Causal Metrics

An Introduction to Regional Impact Evaluation

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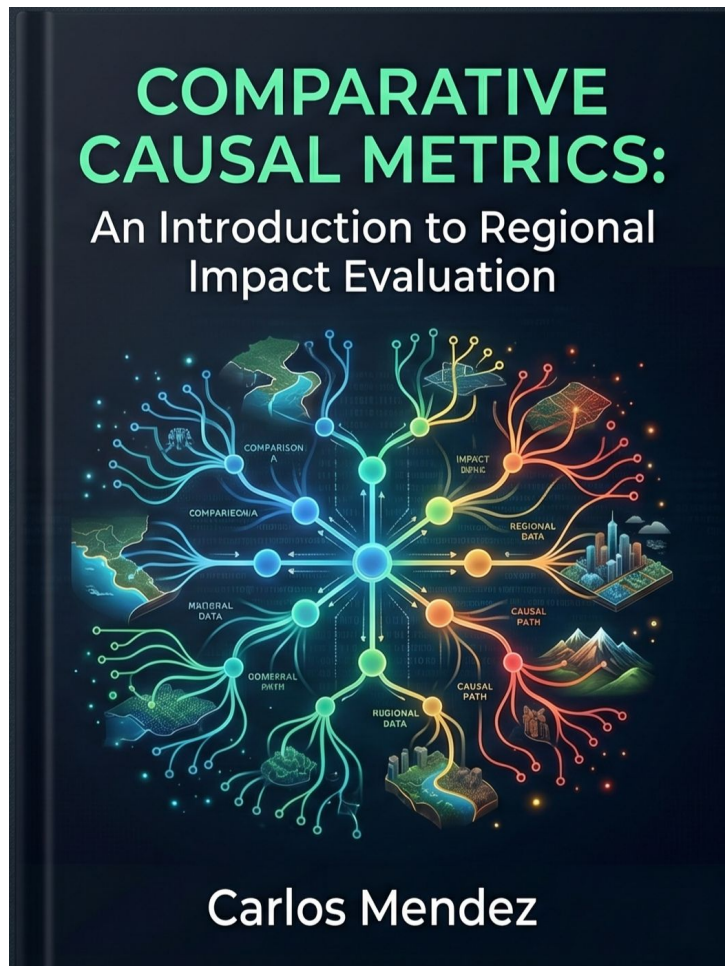
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Preface



0.1 About this book

Comparative Causal Metrics is an introduction to **regional impact evaluation** using modern causal-inference methods implemented entirely in **R** and rendered with **Quarto**. The book is organized around a small set of quasi-experimental techniques that have become workhorses for evaluating the effects of policies, shocks, and interventions on regional outcomes:

- **Interrupted time series** — modeling a single region’s outcome before and after a discrete event.
- **Regression discontinuity in time** — exploiting sharp temporal cutoffs as natural experiments.
- **Classical synthetic control** — constructing a weighted comparison region from a donor pool.
- **Structural Bayesian time series** — uncertainty-aware counterfactual estimation via state-space models.

Each chapter pairs the conceptual logic of the method with a worked R example using publicly available data, so readers can reproduce every estimate end-to-end.

0.2 How to read this book

The chapters are designed to be read sequentially but each is self-contained. Code is folded by default — click `</> Code` in any chapter to reveal the underlying R. The full source is available on [GitHub](#), and PDF / EPUB downloads are linked in the navbar.

0.3 Acknowledgments

This book builds on the infrastructure of the companion project *Mastering Causal Metrics* and on a long tradition of applied econometric scholarship.

1 Introduction

1.1 Overview

Placeholder. This chapter introduces the comparative-causal framework for regional impact evaluation and motivates the four method families covered in the rest of the book.

1.2 What is regional impact evaluation?

The native approach and the need for a comparative-causal framework.

1.3 The potential outcomes framework

Placeholder.

1.4 A roadmap for the book

Placeholder.

1.5 Setup

```
library(tidyverse)
library(fixest)
library(modelsummary)
```

1.6 Worked example: Simple synthetic control

Placeholder. Each subsequent chapter follows this structure: motivation, identifying assumptions, estimation in R, and a regional case study.

1.7 Further reading

Placeholder.

2 Interrupted Time Series

2.1 Overview

Placeholder. Interrupted time series (ITS) designs estimate the effect of a discrete intervention on a single region's outcome trajectory by modeling pre- and post-intervention trends.

2.2 Identifying assumptions

Placeholder.

2.3 Estimation in R

```
library(tidyverse)
library(fixest)
library(segmented)
library(nlme)
```

```
# Placeholder: load a regional time series, fit a segmented regression
# with an intervention dummy and post-trend interaction.
```

2.4 Regional case study

Placeholder.

2.5 Diagnostics

Placeholder.

2.6 Further reading

Placeholder.

3 Regression Discontinuity in Time

3.1 Overview

Placeholder. Regression discontinuity in time (RDiT) extends classical RD designs to settings where the running variable is time itself, with treatment switching on at a known calendar cutoff.

3.2 Identifying assumptions

Placeholder.

3.3 Estimation in R

```
library(tidyverse)
library(fixest)
library(rdrobust)
```

```
# Placeholder: bandwidth selection, local linear regression around the
# cutoff date, robust bias-corrected standard errors.
```

3.4 Bandwidth selection

Placeholder.

3.5 Threats to validity

Placeholder. RDiT designs are vulnerable to time-varying confounders, autocorrelation, and anticipation effects.

3.6 Regional case study

Placeholder.

3.7 Further reading

Placeholder.

4 Classical Synthetic Control

4.1 Overview

Placeholder. The synthetic control method constructs a counterfactual for a single treated region as a convex combination of untreated donor regions, choosing weights to minimize pre-treatment outcome divergence.

4.2 The method

Placeholder.

4.3 Estimation in R

```
library(tidyverse)
library(tidysynth)
```

```
# Placeholder: assemble a donor pool, fit a synthetic control with
# tidysynth, plot treated vs. synthetic trajectories.
```

4.4 Inference: placebo tests

Placeholder.

4.5 Regional case study

Placeholder.

4.6 Limitations

Placeholder. Classical synthetic control is sensitive to donor-pool composition, requires sufficient pre-treatment history, and offers limited frequentist inference.

4.7 Further reading

Placeholder.

5 Structural Bayesian Time Series

5.1 Overview

Placeholder. Structural Bayesian time-series models (e.g., Google’s `CausalImpact`) estimate the causal effect of an intervention as the difference between observed post-intervention outcomes and a Bayesian counterfactual forecast built from pre-intervention dynamics and contemporaneous control series.

5.2 The model

Placeholder. A state-space decomposition into local-trend, seasonal, and regression components, fit by `bsts`.

5.3 Estimation in R

```
library(tidyverse)
library(CausalImpact)
library(bsts)
```

```
# Placeholder: build a CausalImpact input frame with the treated series
# and donor controls, call CausalImpact(), summarize and plot.
```

5.4 Interpreting posterior intervals

Placeholder.

5.5 Regional case study

Placeholder.

5.6 Comparison with synthetic control

Placeholder. `CausalImpact` and classical synthetic control answer related questions with different assumptions; this section compares estimates on the same data.

5.7 Further reading

Placeholder.

References